## Lesson Objectives

In this lesson I will introduce three important concepts and give a quick overview of E-tivity 2.

## Transformation of feature spaces

In this E-tivity we will start with a look at feature transformations. These are used extensively in machine learning to represent the data provided to a machine learning model in such a way, that the machine learning model can learn the trends of interest as easily as possible.

We will demonstrate these feature transformations in this E-tivity by using linear regression, which results in linear predictions in the feature space, for the modelling of a non-linear trend. We will see that this becomes possible by adding higher order powers of the feature in the feature vector. As a result, trends which we could not model with any reasonable accuracy previously, can now be modelled with reasonable accuracy.

An important aside is that it may be tempting to choose these features based on the trends you observe in the data. However, if you do this, you inject yourself in the machine learning algorithm which means you may be engaging in so-called data snooping. As a result of such data snooping, the guarantee provided by the VC bound will no longer hold, unless you charge a higher VC dimension. We will see what this means during the E-tivity.

## Finding the right complexity

Once we have used non-linear transformations to find a reasonable model for the data in-sample, we will start looking at the out-of-sample performance again, and from E-tivity 1 we know that it is key to limit the complexity of our model to get good out-of-sample performance.

We will look at two ways of optimising the out-of-sample performance.

Regularisation is the first one and is a tremendously important concept in any machine learning workflow. Regularisers can often be considered as hyper parameters of your model. Such hyper parameters are not updated as part of the learning algorithm, but are controlled by you.

We will focus on regularisers that add a penalty to the error function. The result of this penalty generally is that weights are encouraged to remain small or even zero. And this in turn ensures that the model's complexity is kept in check, and with that hopefully the generalisation error of the model.

Cross validation is the second technique extensively used to find the optimal hyper parameters for a model. In cross validation, you train the model on one dataset, and then estimate its out of sample performance on another dataset. Now, we already know that out-of-sample data is per definition not available to us. However, we can of course put part of our training data separate when we are training and then use this data to test the trained model. As the model has not seen this data, it can be considered out of sample. Doing this of course reduces the amount of data available for training, which we now know reduces our ability to use a model with high complexity. But we will see that cross validation minimises the impact of this reduction in available training data and allows use to use all training for validation as well as for testing.

## Logistic Regression

Having looked at linear regression in quite some detail in the previous E-tivity, we will now look at logistic regression in more detail. We will see that logistic regression uses the features and the class membership of these features to learn a probability that a given set of features belongs to a certain class. Logistic Regression uses the log-loss error function and we will have a look at this error function in some more detail. As always, you do not need to be able to replicate the mathematical derivations we will present, but it is important to build an intuitive understanding of this important model.

## Purpose Task 1 & 2

The purpose of Task 1 and 2 is to try non-linear transformations of the original feature to allow a linear algorithm to do well on a non-linear task. In task 1 you will look at a classification problem and in task 2 you will tackle regression.

Task 3 will introduce regularisation as a means of controlling the expressiveness of your model, that we have just increased in Task 1 and 2, to a level that is most appropriate for the task in hand.

We encourage you to upload your work on these tasks in time and use the E-tivity forum to discuss these concepts with your peers.

## Purpose Task 4

Task 4 is a smaller task in which you will see the power of cross validation in combination with regularisation.

We expect you will not have time to start on Task 4 in the first week and hence there is no intermediate deadline for this task. However, of course this does not mean that you should not feel free to discuss this task in the E-tivity forum.